



Problem Statement:

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  void findPair(int arr[], int low, int high, int x) {
4      if (low >= high) {
5          printf("No\n");
6          return;
7      }
8
9      int sum = arr[low] + arr[high];
10     if (sum == x) {
11         printf("%d\n%d\n", arr[low], arr[high]);
12         return;
13     }
14
15     if (sum < x) {
16         findPair(arr, low + 1, high, x);
17     } else {
18         findPair(arr, low, high - 1, x);
19     }
20 }
21
22 int main() {
23     int n;
24     scanf("%d", &n);
25
26     int arr[n];
27     for (int i = 0; i < n; i++) {
28         scanf("%d", &arr[i]);
29     }
30
31     int x;
32     scanf("%d", &x);
33
34     findPair(arr, 0, n - 1, x);
35
36     return 0;
37 }
```

Check

	Input	Expected	Got	
✓	4	4	4	✓
	2	10	10	
	4			
	8			
	10			
	14			

	Input	Expected	Got	
✓	5	No	No	✓
	2			
	4			
	6			
	8			
	10			
	100			

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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